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Vapor Stone Rail Systems

Generic Software Documentation Project

Project # 7260

Software Design Description (SDD)
Door Operator Module

Contract Number		CDRL Number	
		Customer Ref. Number	

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Preface

This Software Design Description (SDD) complies with the requirements of the IEEE 1016-1998 and 1558-2004.

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1 INTRODUCTION

This document is the Software Design Description of the Door Operator (DOP) Module developed by Vapor Stone Rail Systems.

The DOP Module is a part of the Door Control Module (DCM) Application that manages the door command and controls the behavior of the motor.

1.1 Purpose

The purpose of this Software Design Description (SDD) is to describe the software components and implementation required details of the DOP Module.

The intended audience is the Vapor Stone Rail Systems software development team and maintenance team members of the transit authority and the car builder.

1.2 Scope

The DOP Module is a software module of the Door Control Module (DCM) Application. It is responsible for managing the door command.
This module interfaces with other modules and, it is responsible for a door operation.

1.3 Definitions and Acronyms

1.3.1 Definitions

Door Control Module: A specific processor based electronic module running a Door Control Module Application. It is used for controlling the door motor.

1.3.2 Acronyms

Acronym	Meaning
ASCII	American Standard Code for Information Interchange
CDRL	Contract Data Requirements List
CPU	Central Processing Unit
CRC	Cyclic Redundancy Check
DCM	Door Control Module
DOP	Door Operator
ECD	Engineering Change Document
IEEE	Institute of Electrical and Electronics Engineers
IO	Input / Output
KB	Kilobytes
LED	Light Emitting Diode
LLC	Limited Liability Corporation
MHz	MegaHertz
MONIT	Monitoring
ms	millisecond
NVRAM	Non-Volatile Random Access Memory
OS	Operating System
PID	Proportional Integral Derivative
PSS	Panel Sensing Switch
PT	Portable Test
PTA	Portable Test Application
PWM	Pulse Width Modulation
RAM	Random Access Memory

Acronym	Meaning
SCI	Software Configuration Items
SDD	Software Design Description
SH	Seek-Home
SRAM	Static Random Access Memory
SRS	Software Requirements Specification
Std	Standard
TS	Technical Specification
VSRS	Vapor Stone Rail Systems

1.4 Document Organization

This SDD in divided into three sections and the following specific information is provided within each one:

- Section 1, Introduction, states the purpose, identifies the scope, gives definitions, abbreviations and an overview of the SDD.
- Section 2, References, lists the factors that affect the product and its requirements.
- Section 3, Software Design, summarizes followings:
    - Decomposition Description: describes functions of software entities.
    - Dependency Description: gives a brief description of the dependency between each entity.
    - Interface Description: describes the interaction between entities.
    - Detailed Design: details the functions implemented by each software entity, presents and explains tasks performed by each software entity in functional conditions.

Within the document, diagrams, flowcharts and tables are used to present the DOP Module design:

- Decision flowcharts give a graphical view of how entities and functions process work.
- The diagram Figure 3-1 describes the dependency between different modules.
- Tables contain data structures used by the DOP Module. They also present C code.

2 REFERENCES

This section lists the number, title, revision, date, and source of all documents referenced in this SDD.

Standards

IEEE Std 1016-1998	IEEE Recommended Practice for Software Design Descriptions, September 23rd 1998, IEEE, www.ieee.org
IEEE Std 1558-2004	IEEE Recommended Practice for Software Documentation for Rail Equipment and Systems, March 23rd 2005, IEEE, www.ieee.org

Vapor Stone Rail Systems Documents

S17-07	Software Design Description Template, Rev. 0, 2002-01-15

3 SOFTWARE DESIGN

3.1 Decomposition Description

3.1.1 Module Decomposition

3.1.1.1 Door Operator Module Description

The DOP Module is composed of the following functions:

- Manage the door command
- Control the behavior of the door
- Manage the door configuration parameters
- Test the motor drive transistor

3.1.2 Concurrent Process Decomposition

3.1.2.1 Door Operator Task Description

The DOP task loop waits for the mail to arrive in the OS mail box before calling the appropriate internal functions. It also calls the Watchdog_Trig function every loop.

3.1.3 Data Decomposition

3.1.3.1 Door Operator Data Description

The DOP Module uses the following structure to control the motor.

Table 3-1: ConfigParametersStruct data bytes definition

Identification	ConfigParameters
Entity Type	Structure ConfigParametersStruct

Prototype	`typedef struct _ConfigPar`
	`{`
	`  unsigned int  ucOpeningTime;  // Opening time.`
	`  unsigned int  ucClosingTime;  // Closing time.`
	`  unsigned char  ucObsRetries;  // Number of retries upon obstruction.`
	`  unsigned char  ucSOSDelay;  // Secondary obstruction delay.`
	`  unsigned char  ucObsRange;  // Obstruction detection range.`
	`  unsigned char  ucObsRopnDist;  // Reopen distance upon obstruction.`
	`  unsigned char  ucObsRopnDly;  // Obstruction Re-open delay.`
	`  unsigned char  ucObsRclsDly;  // Obstruction Re-close delay.`
	`  unsigned int  ucDlayOpen;  // Delay before starting to open.`
	`  unsigned int  ucDlayClose;  // Delay before starting to close.`
	`  unsigned char  ucFlashRate;  // Flashing rate for light or audio.`
	`  unsigned char  ucDummyByte;  // Padding byte for aligning it even.`
	`}ConfigPar;`
	
	`Typedef struct ConfigParametersStruct`
	`{`
	`  ConfigPar  Data;`
	`  unsigned int  uiCRC16;`
	`}ConfigParametersStruct;`
	
	`ConfigParametersStruct far ConfigParameters;`

3.2 Dependency Description

3.2.1 Inter-Module Dependencies

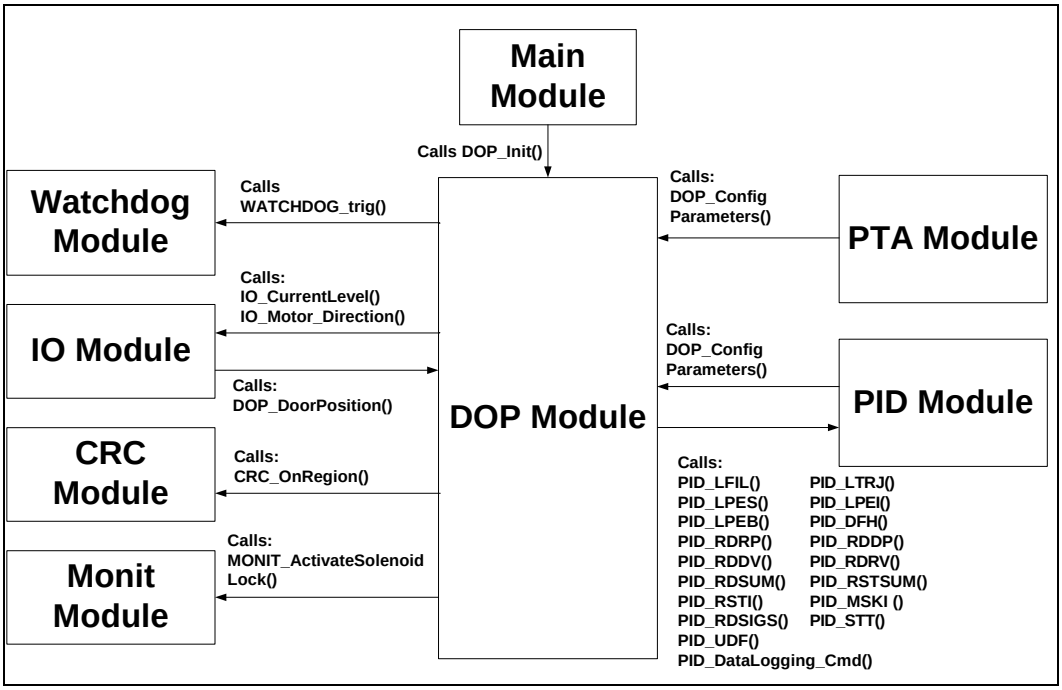

Figure 3-1: Inter-Module Dependencies Diagram

3.2.1.1 Door Operator Module

Figure 3-1 shows the Inter-Module Dependencies Diagram of the DOP module.

3.2.2 Inter-Process Dependencies

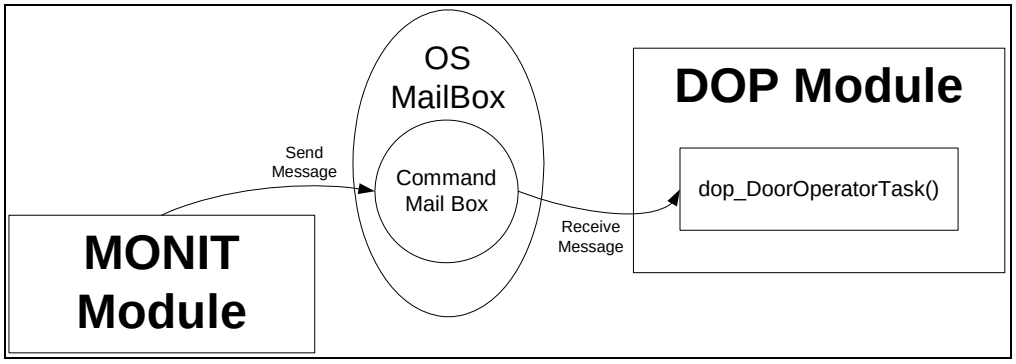

Figure 3-2: Inter-Process Dependencies Diagram

As seen in Figure 3-2, the dop_DoorOperatorTask function of the DOP Module waits for the message from the MONIT Module for the door operation command.

3.2.3 Data Dependencies

DOP Module is dependant to the SignalsIn and CPU_SignalsOut of the IO Module.

3.3 Interface Description

This section describes the functions of DOP module in alphabetical order.

3.3.1 Module Interfaces

3.3.1.1 DOP Module External Interfaces

Function Name	unsigned int DOP_ConfigParameters (unsigned int uiParameterCode, unsigned int uiValue)
Description	This function validates the new value and updates it to the ConfigParameters.
Input	uiParameterCode: The parameter code. uiValue: The new value to update.
Output	None.
Return	The newly assigned or present value of the parameter code.

Function Name	void DOP_Init (void)
Description	This function initializes the ConfigParameters data and calls the DOP_DoorOperatorTask function.
Input	None.
Output	None.
Return	None.

3.3.1.2 DOP Module Internal Interface

Function Name	void dop_Close_Seq (void)
Description	This function moves the door to the fully closed position.
Input	None.
Output	None.
Return	None.

Function Name	void dop_Closed_Locked_Seq (void)
Description	This function resets obstruction counter and turns off the motor.
Input	None.
Output	None.
Return	None.

Function Name	void dop_Current_Limit_Locking_Testing (void)
Description	This function tests the current limit circuit by verifying the current while increasing the PWM. This is only done by compressing the seals to each other.
Input	None.
Output	None.
Return	None.

Function Name	void dop_DoorOperatorTask (void)
Description	This function waits for a message to arrive from the OS mail box, and then it calls the appropriate function according to the message.
Input	None.
Output	None.
Return	None.

Function Name	void dop_Emergency_Handle_Seq (void)
Description	This function stops motor and wait for the next command.
Input	None.
Output	None.
Return	None.

Function Name	void dop_Fully_Open_Seq(void)
Description	This function maintains the door to fully open position.
Input	None.
Output	None.
Return	None.

Function Name	void dop_Fully_Stop_Seq(void)
Description	This function resets the obstruction counter.
Input	None.
Output	None.
Return	None.

Function Name	void dop_Keep_Driving_Seq(void)
Description	This function keeps driving the door to the close position while applying small force.
Input	None.
Output	None.
Return	None.

Function Name	void dop_Keep_Driving_Obstruction_Seq(void)
Description	This function keeps driving the door to the close position with defined (constant value) time.
Input	None.
Output	None.
Return	None.

Function Name	void dop_Local_Recycle_Seq(void)
Description	This function re-opens the door and sets the door state to “Close Door” after.
Input	None.
Output	None.
Return	None.

Function Name	unsigned char dop_MotorDriveTest(void)
Description	This function tests the motor drive circuit by activating the transistor one at a time.
Input	None.
Output	None.
Return	None.

Function Name	void dop_Open_Seq (void)
Description	This function moves the door to fully open position.
Input	None.
Output	None.
Return	None.

Function Name	void dop_Recycle_Seq (void)
Description	This function re-opens the doors whenever closing obstruction occurs.
Input	None.
Output	None.
Return	None.

Function Name	void dop_SeekHome_Close_Seq (void)
Description	This function moves the door to close direction with a low speed.
Input	None.
Output	None.
Return	None.

Function Name	void dop_SeekHome_Closing_Obstruction_Seq (void)
Description	This function moves the door in closing direction in every second until the obstruction retries is expired.
Input	None.
Output	None.
Return	None.

Function Name	void dop_SeekHome_Door_Stop_Seq (void)
Description	This function stops the motor.
Input	None.
Output	None.
Return	None.

Function Name	void dop_StartUp (void)
Description	This function makes sure the motor is off and resets the obstruction counter.
Input	None.
Output	None.
Return	None.

Function Name	void dop_Stop_To_Close_Seq (void)
Description	This function first stops the door, and then it moves the door to the fully closed position.
Input	None.
Output	None.
Return	None.

Function Name	void dop_Unlock_Seq (void)
Description	This function compresses the seals and unlocks the door.
Input	None.
Output	None.
Return	None.

3.4 Detailed Design

This section describes how the Door Operator Module works and what operation it performs.

3.4.1 Door Operator Module Detailed Design

This module implements functions to control the motor and the solenoid behavior. The following sub-sections describe each function.

The event code numbers for the followings are defined in the *SACS* document. (Please see Appendix A)

- Motor OC Shutdown Circuit Failure
- Door Opening Time Longer than Expected
- Door Closing Time Longer than Expected
- Door Locking Time Longer than Expected
- Normal Current Limit Error Detected
- High Current Limit Error Detected
- Door Closing Obstruction Detected
- Door Opening Timeout
- Motor Control Module Communication Error
- Current Level Selector Error
- Motor Drive Circuit (H-Bridge) Error
- Motor Rotation with Door Locked

3.4.1.1 dop_DoorOperatorTask Function

This task function first waits for the door command from other modules and then interprets the command accordingly to the present condition in the “Command Interpretation”. The “State Parser” then parses the state to call the right function.

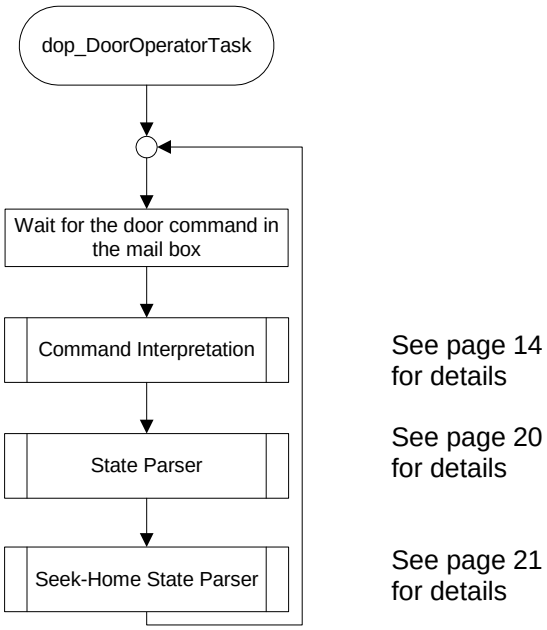

Figure 3-3: DOP_DoorOperatorTask

3.4.1.1.1 Command Interpretation

Figure 3-4 shows the processing done by the task when a command is received.

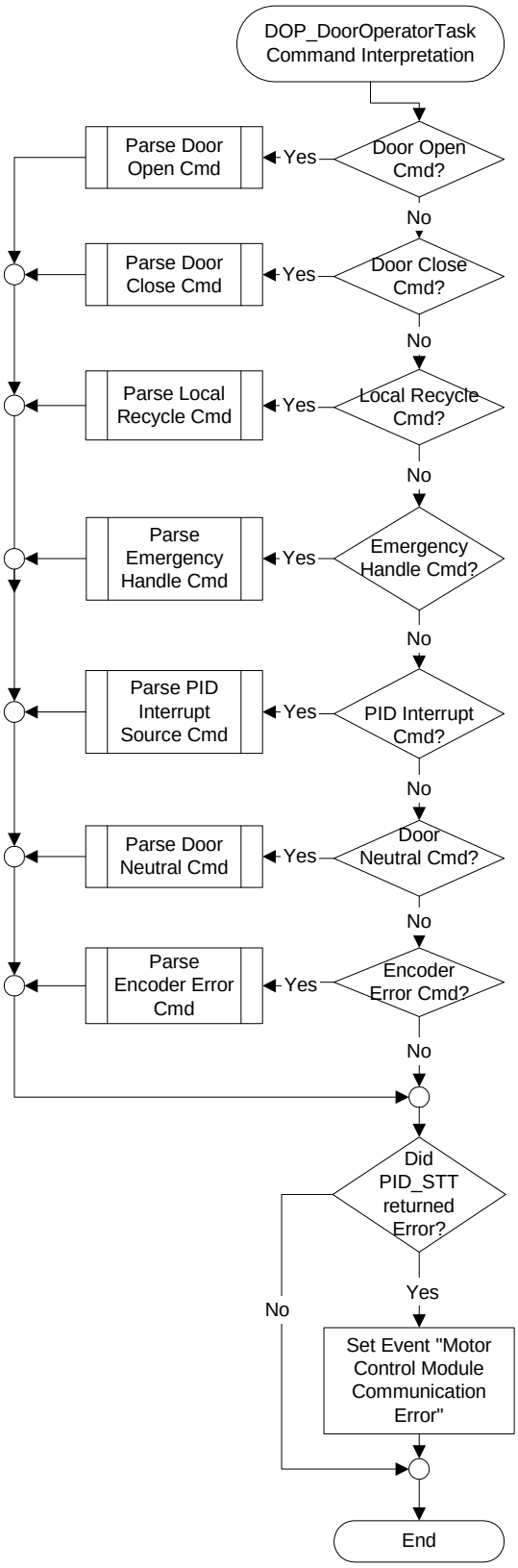

Figure 3-4: DOP_DoorOperatorTask Command Interpretation

a) Parse Door Open Command:

Figure 3-5 describes the process of parsing a door open command.

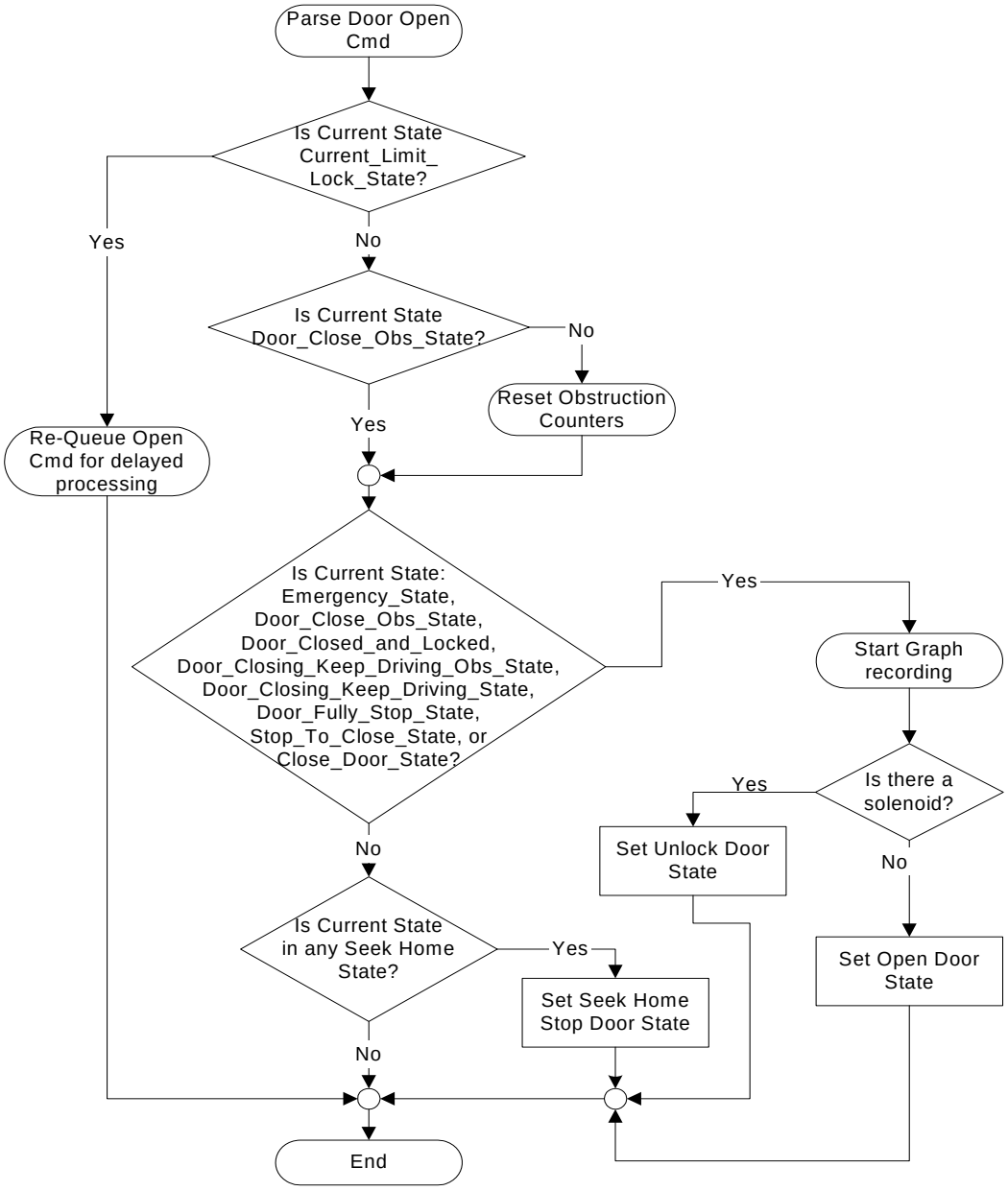

Figure 3-5: Parse Door Open Command (Command #1)

b) Parse Door Close Command:

Figure 3-6 describes the process of parsing a door close command.

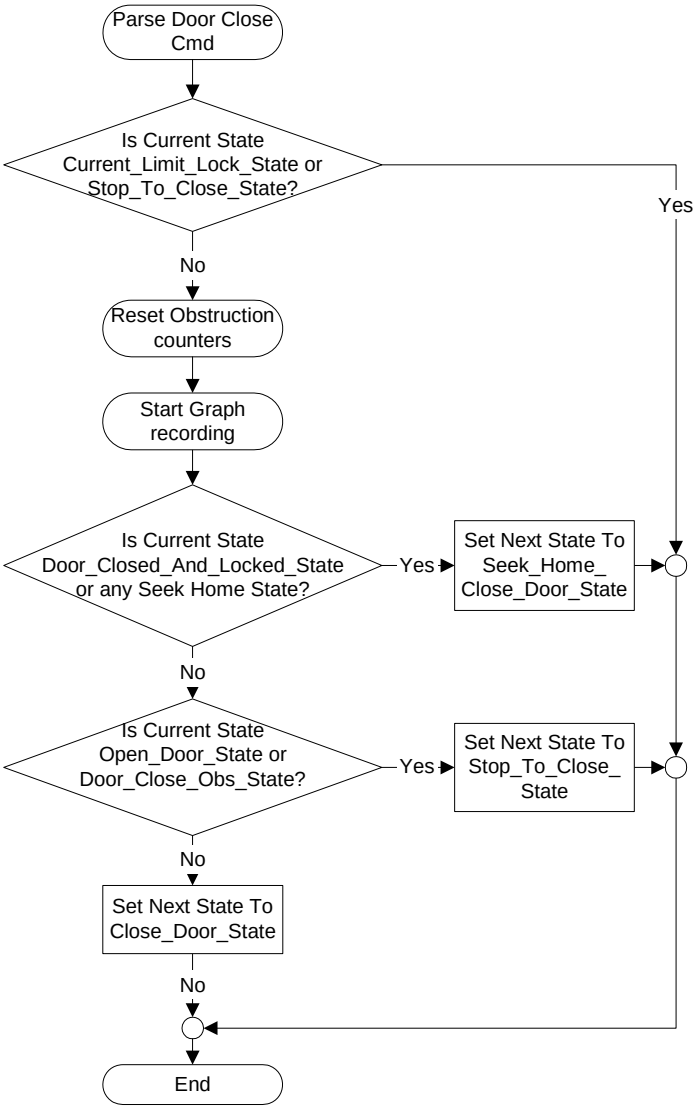

Figure 3-6: Parse Door Close Command (Command #2)

c) Parse Local Recycle Command:

Figure 3-7 describes the process of parsing a local recycle command.

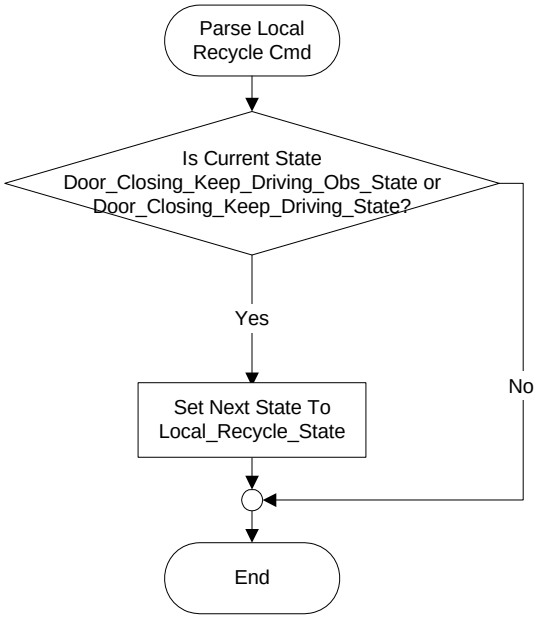

Figure 3-7: Parse Local Recycle Command (Command #3)

d) Parse Emergency Handle Command:

Figure 3-8 describes the process of parsing an emergency handle command.

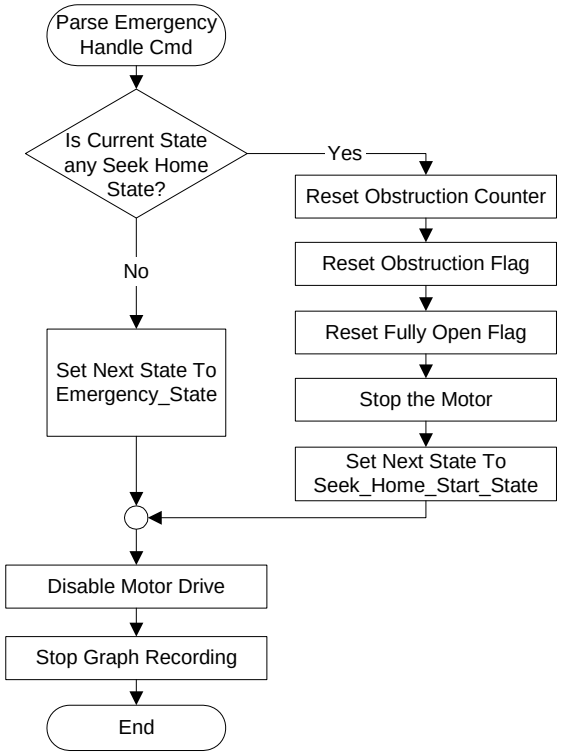

Figure 3-8: Parse Emergency Handle Command (Command #4)

e) Parse PID Interrupt Command:

Figure 3-9 describes the process of parsing a PID interrupt command. The interrupt source is verified and only the interrupt from an excessive position error is processed.

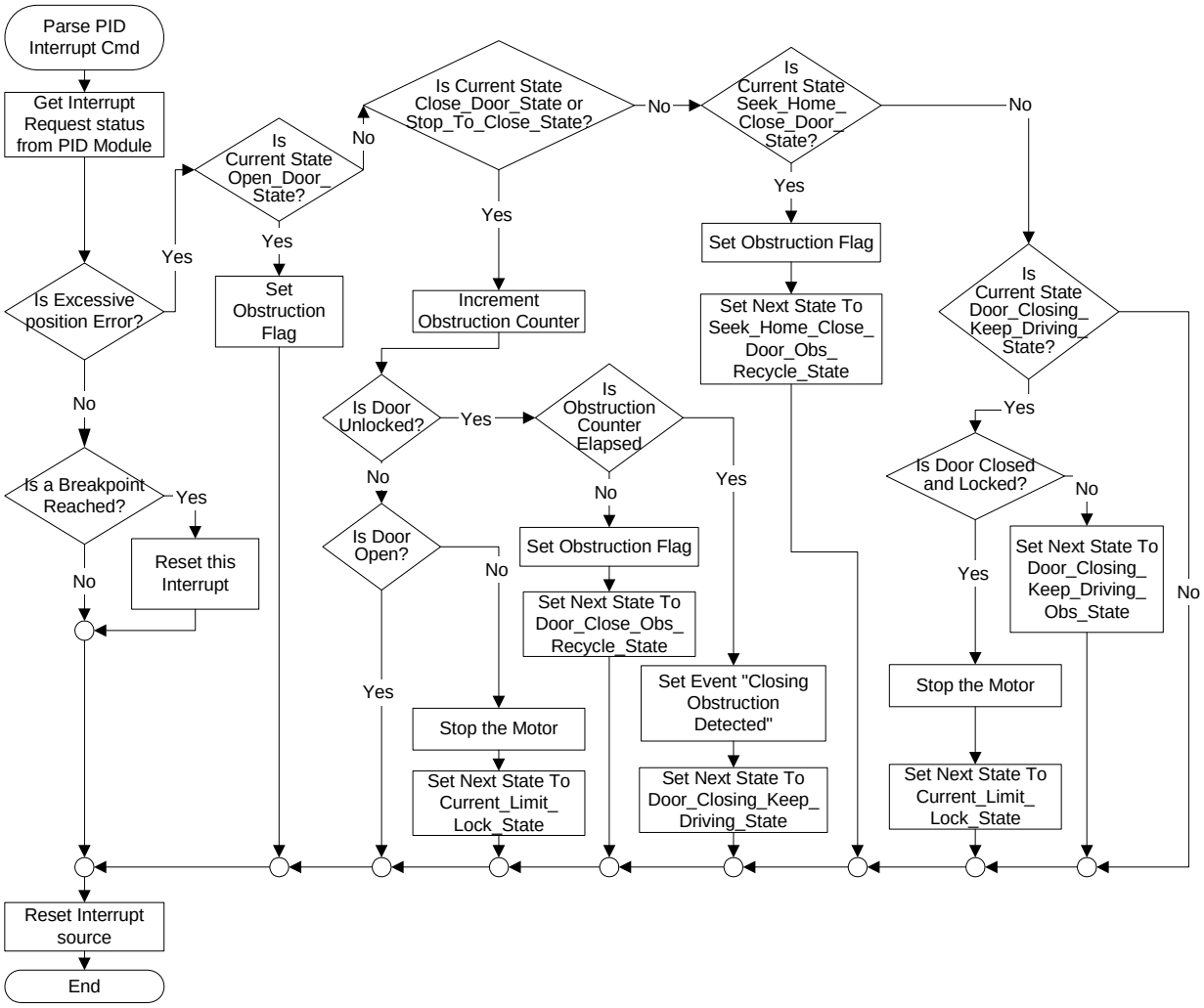

Figure 3-9: Parse PID Interrupt Command (Command #5)

f) Parse Door Neutral Command:

Figure 3-10 describes the process of parsing a door neutral command.

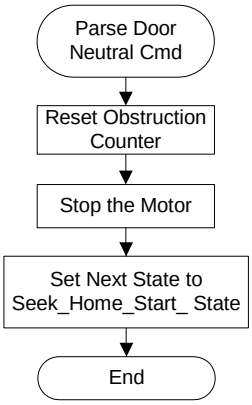

Figure 3-10: Parse Door Neutral Command (Command #6)

g) Parse Encoder Error Command:

Figure 3-11 describes the process of parsing an encoder error command.

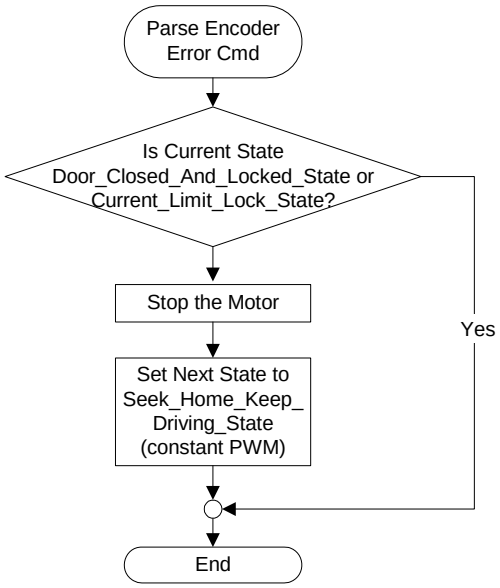

Figure 3-11: Parse Encoder Error Command (Command #7)

3.4.1.1.2 State Parser

Figure 3-12 describes the parsing done for each of the state set by the Command Interpreter.

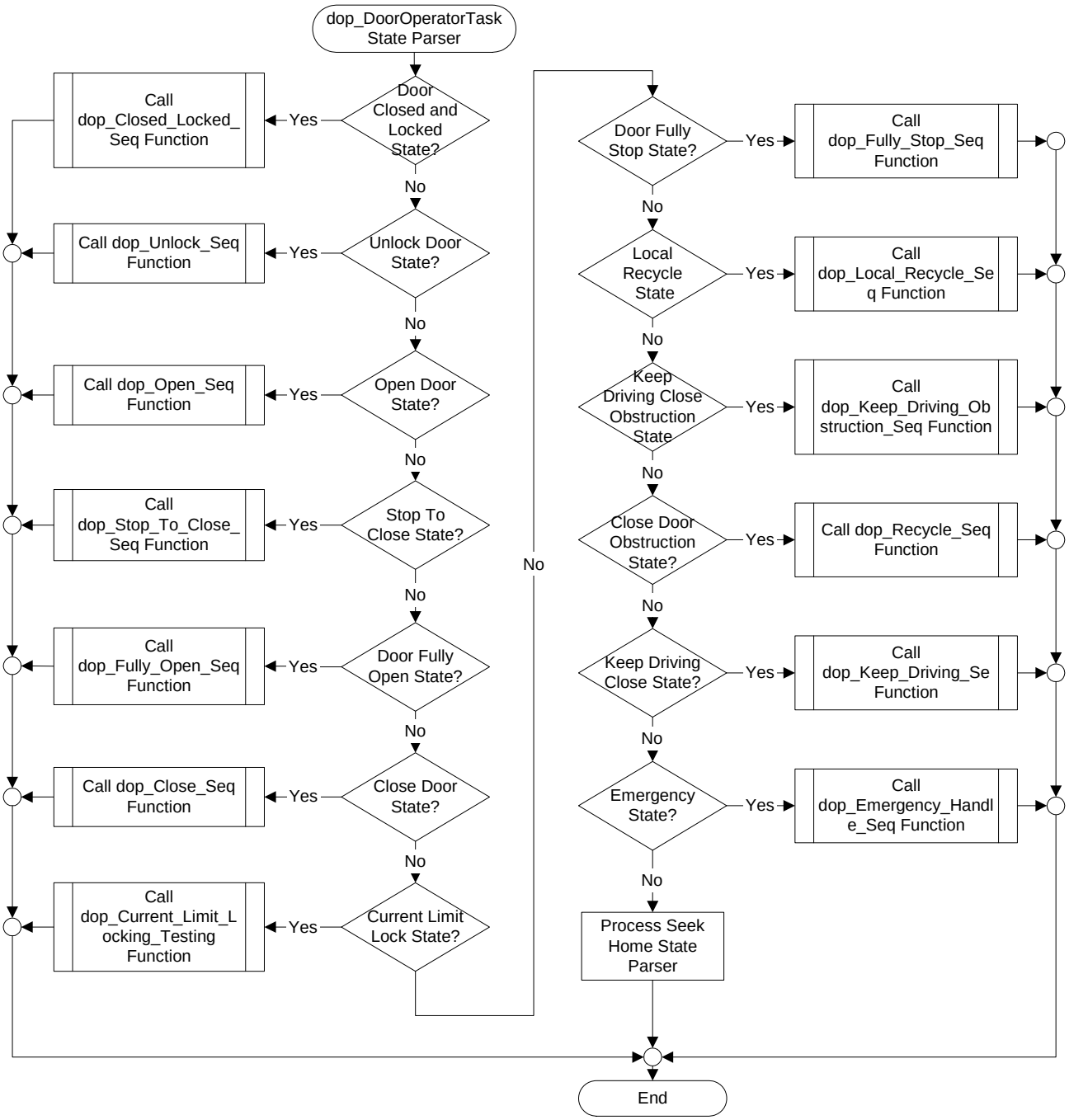

Figure 3-12: dop_DoorOperatorTask State Parser

3.4.1.1.3 Seek-Home State Parser

If this module is running for the first time (like after a reset or a power failure), and it receives a Door Close command from the Monitoring module, it executes the Door Seek-Home Close Operation state diagram described in Figure 3-23. The Door Operator State Parser shown in Figure 3-12 parses the Door Close command and calls the Door Close Parser shown in Figure 3-6 and sets the state to Start-Up. This is the first state of the Door Seek-Home Operation. Figure 3-13 shows the Door Operator Seek-Home State Parser and describes all the door operations when in Seek-Home mode. Figure 3-22 shows the sequence used to perform a Start-Up during a Door Seek-Home Operation.

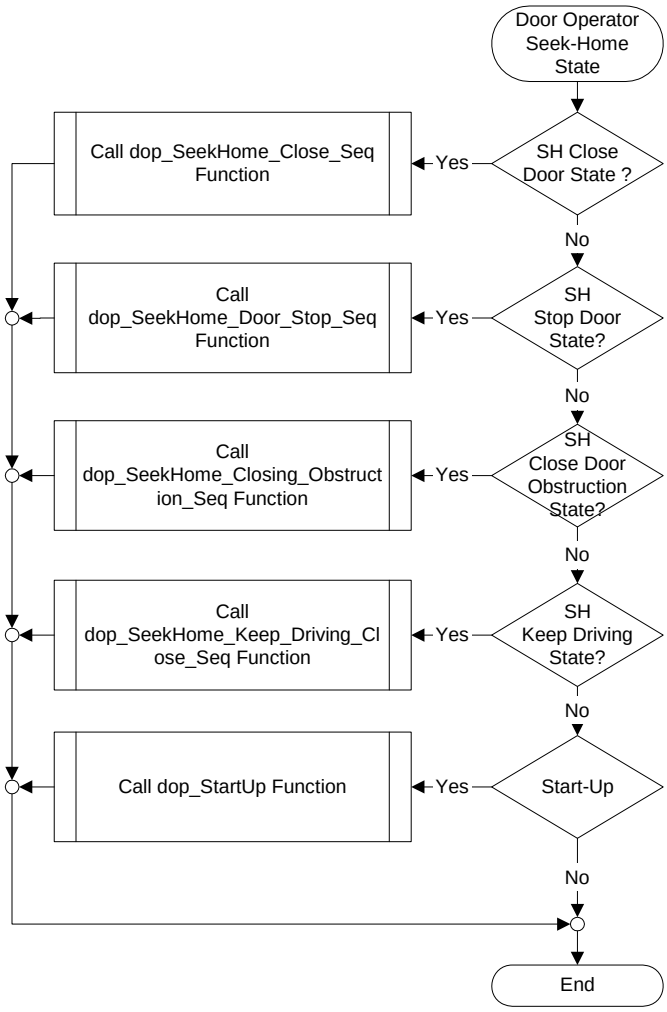

Figure 3-13: Door Operator Seek-Home State Parser

3.4.1.2 dop_Unlock_Seq Function

Figure 3-14 describes the sequence used to perform a Door Unlock.

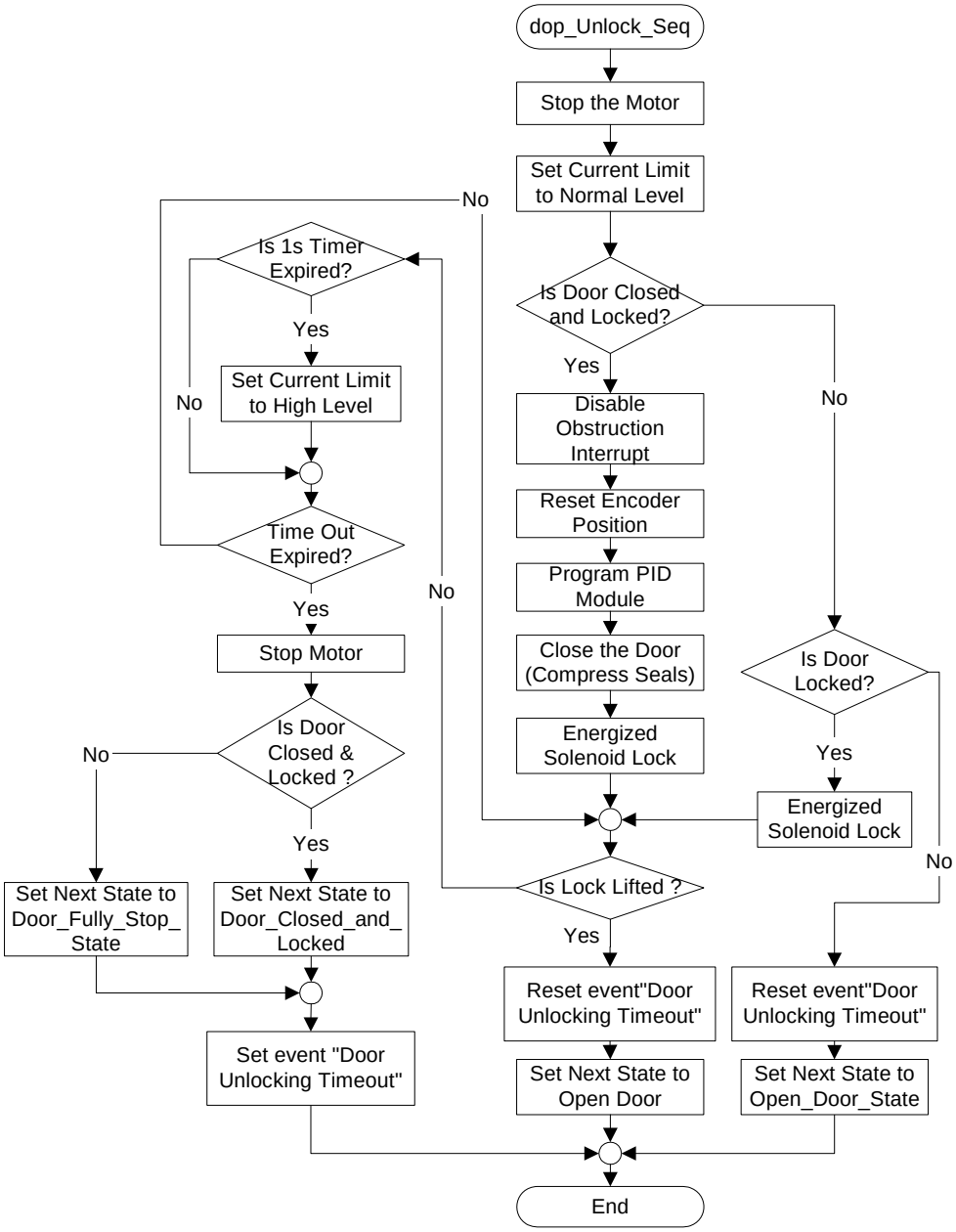

Figure 3-14: dop_Unlock_Seq Flowchart

3.4.1.3 dop_Open_Seq Function

Figure 3-15 shows the sequence used to perform Door Open.

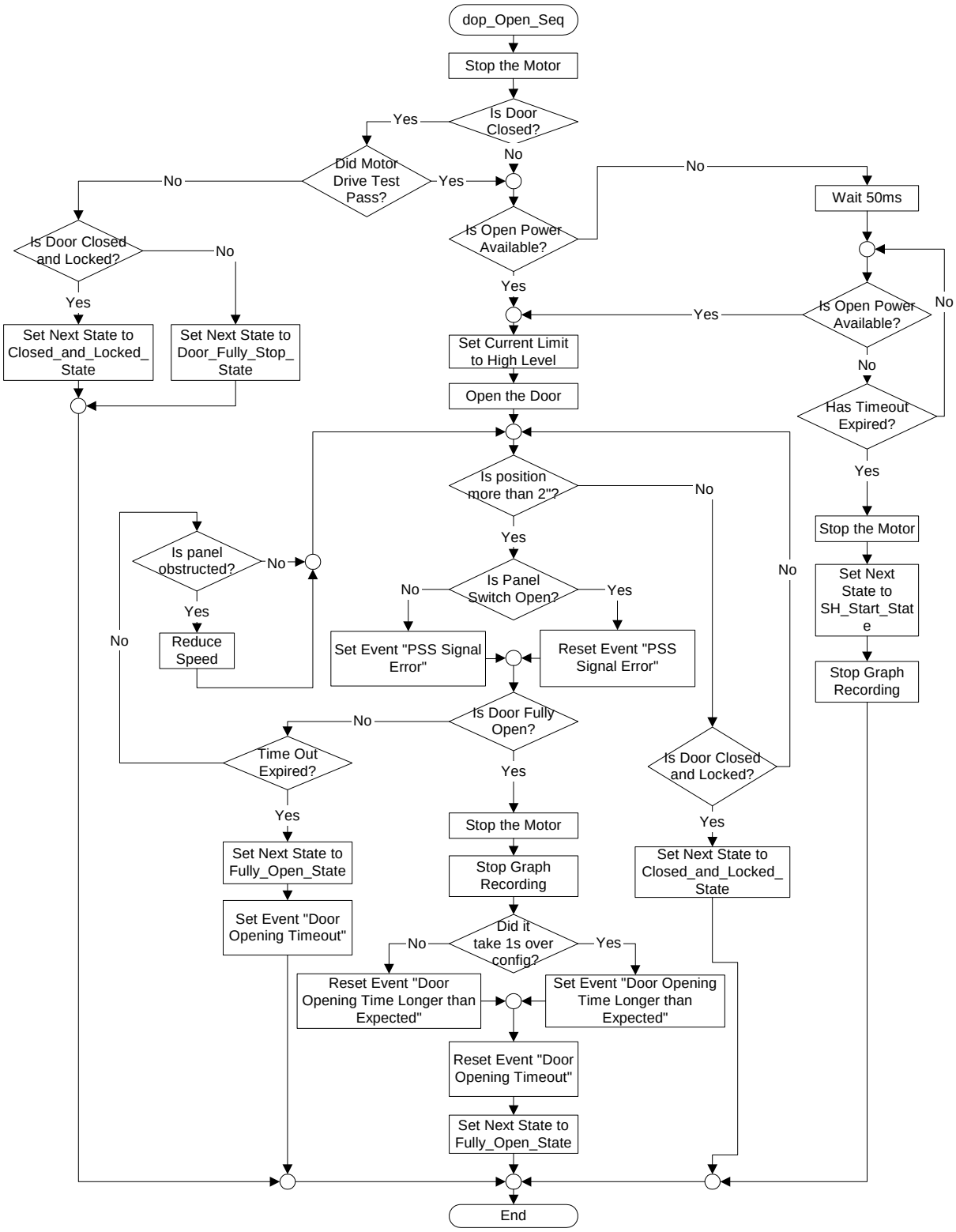

Figure 3-15: dop_Open_Seq Flowchart

3.4.1.4 dop_Fully_Open_Seq Function

Figure 3-16 shows the sequence used to perform Door Fully Open.

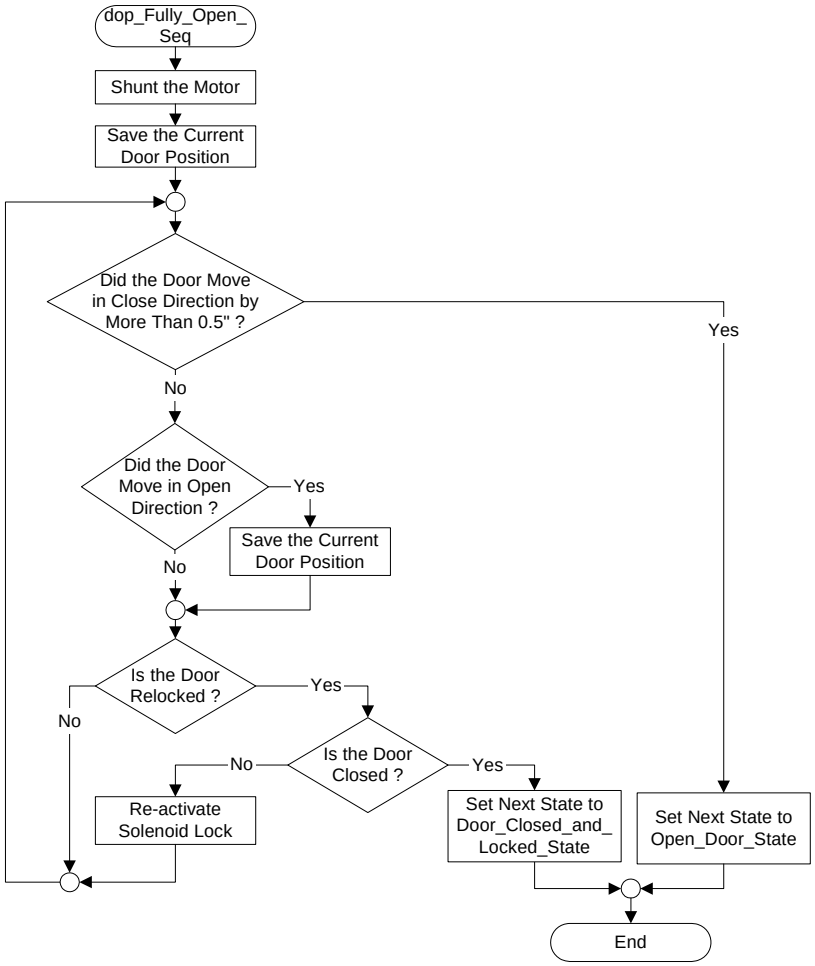

Figure 3-16: dop_Fully_Open_Seq Flowchart

3.4.1.5 dop_Fully_Stop_Seq Function

This function resets the obstruction counter.

3.4.1.6 dop_Stop_To_Close_Seq Function

Figure 3-17 shows the sequence used to perform a Door Stop to Close.

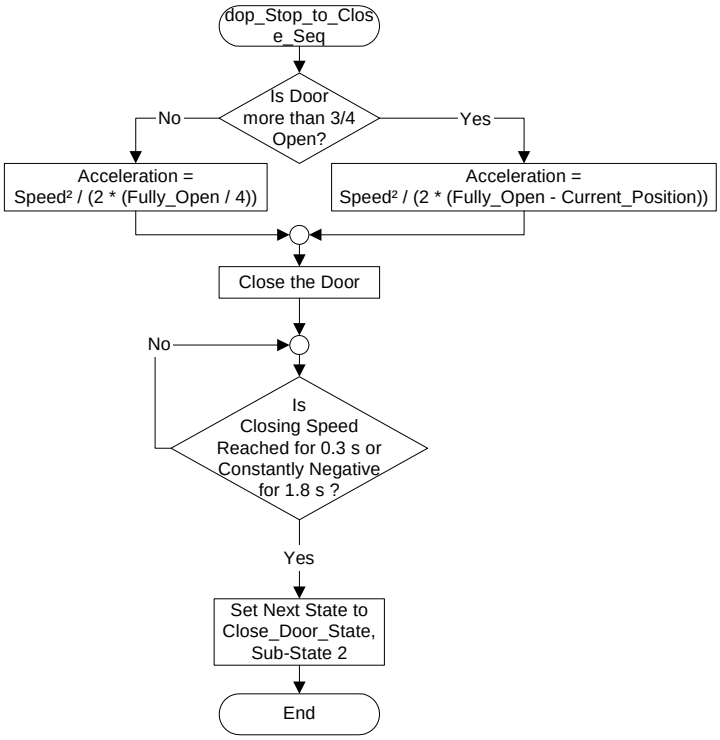

Figure 3-17: dop_Stop_To_Close_Seq Flowchart

3.4.1.7 dop_Close_Seq Function

Figure 3-18 describes the sequence used to perform a Door Close.

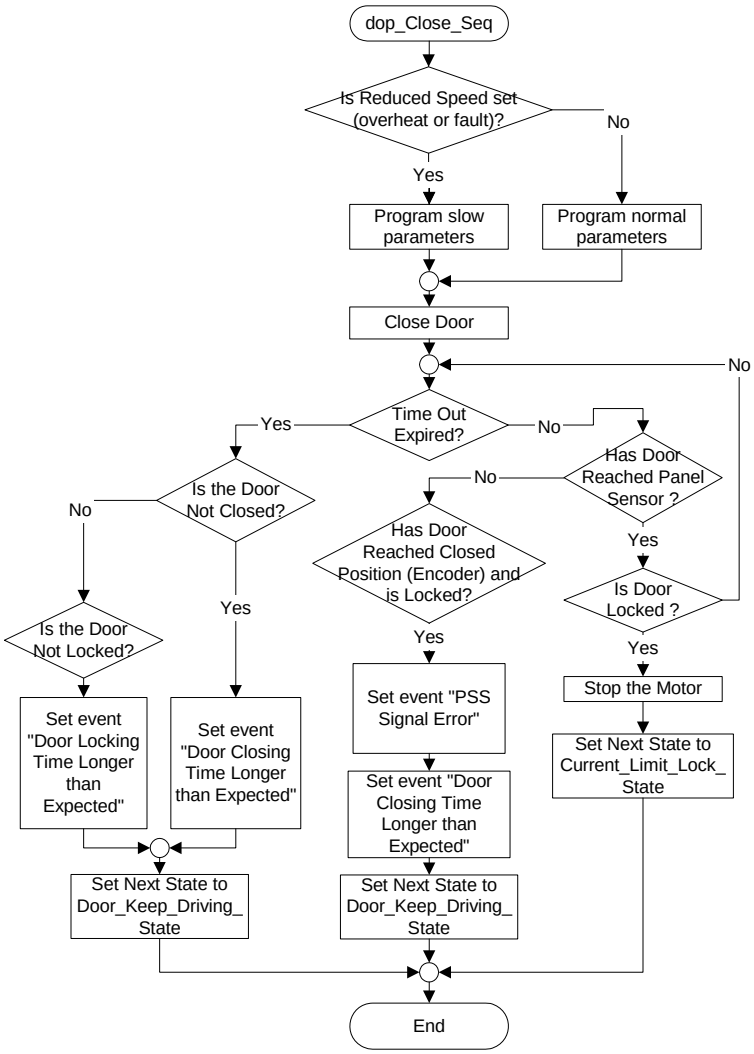

Figure 3-18: dop_Close_Seq Flowchart

3.4.1.8 dop_Current_Limit_Locking_Testing Function

Figure 3-19 shows the sequence used to perform a Current Limit Lock sequence.

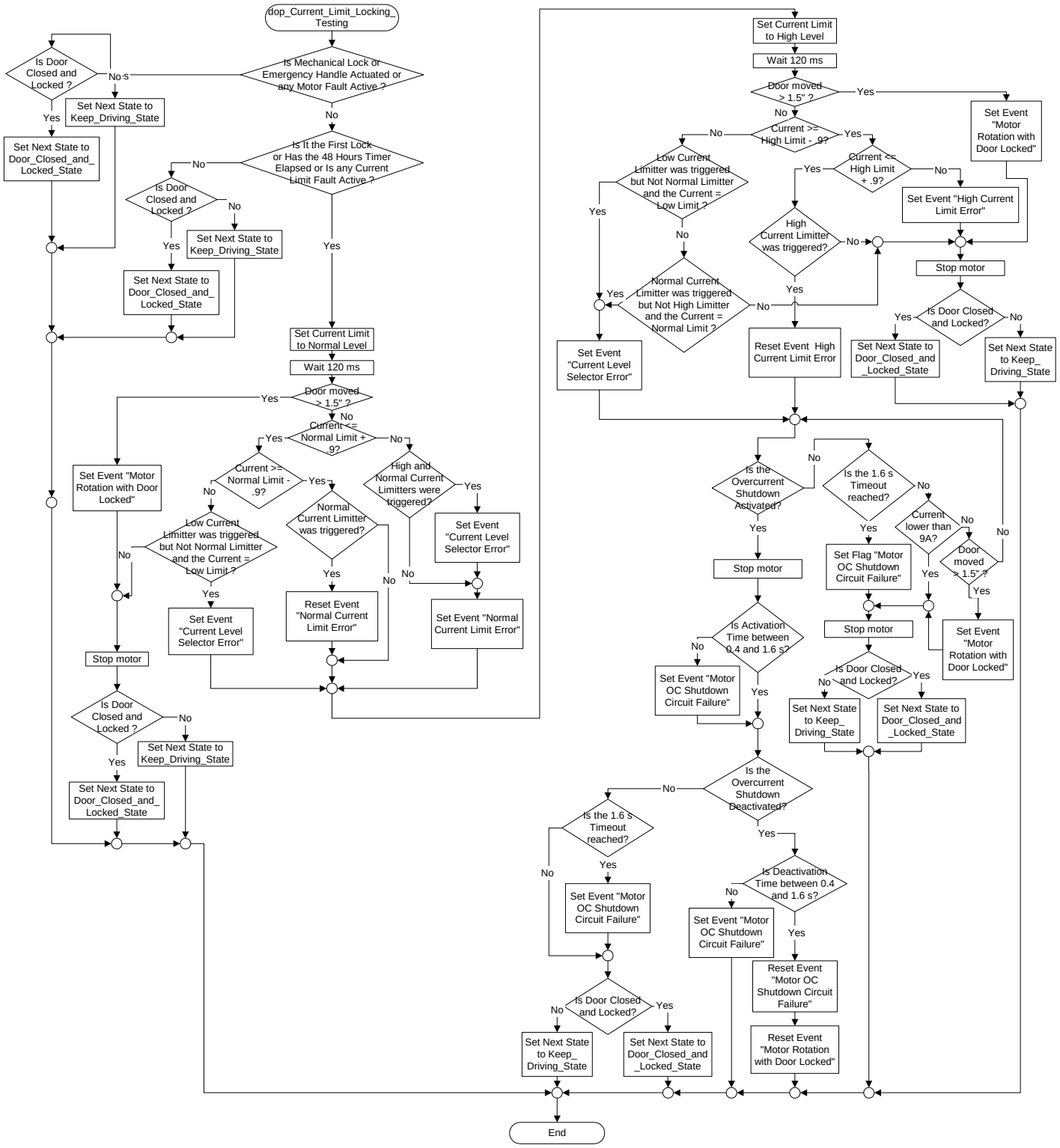

Figure 3-19: dop_Current_Limit_Locking_Testing Flowchart

3.4.1.9 dop_Local_Recycle_Seq Function

Figure 3-15 shows the sequence used to perform Local Recycle. The Local Recycle is used to retry to close obstructed doors. The door is re-opened then jumps to the Door Close sequence.

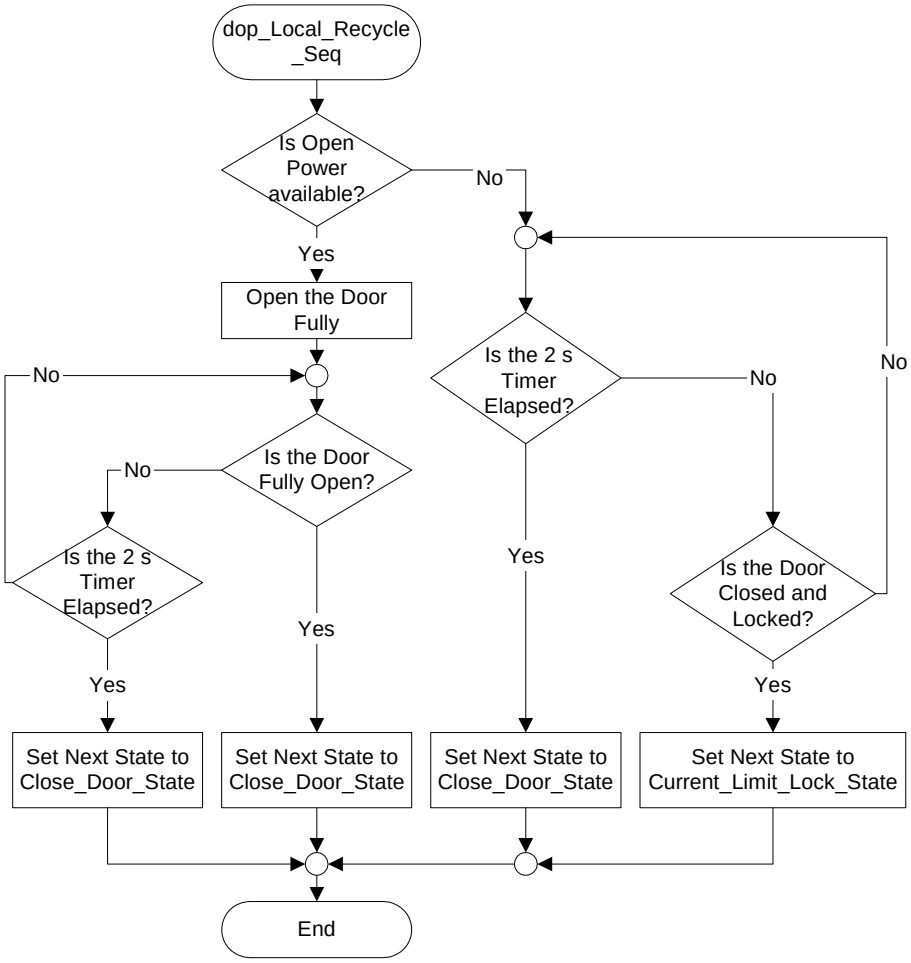

Figure 3-20: dop_Local_Recycle_Seq Flowchart

3.4.1.10 dop_Recycle_Seq Function

Figure 3-21 shows the sequence used to perform a Door Closing Obstruction.

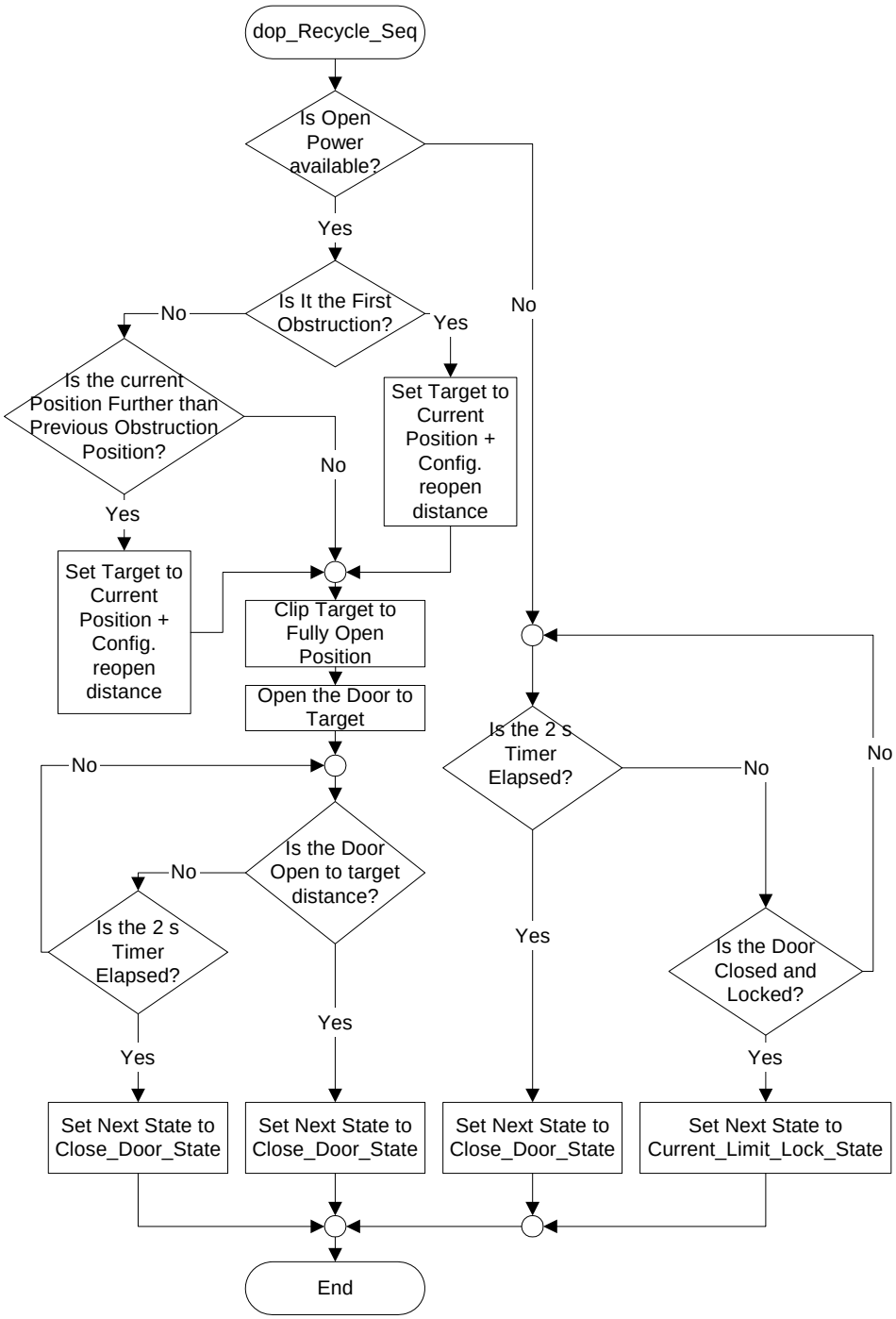

Figure 3-21: dop_Recycle_Seq Flowchart

3.4.1.11 dop_Keep_Driving_Seq Function

The Keep Driving Sequence is identical to the Door Close sequence, seen in Figure 3-18, except that the obstruction detection is disabled.

3.4.1.12 dop_Keep_Driving_Obstruction_Seq Function

This function is identical to the dop_Keep_Driving_Seq Function (Section 3.4.1.11) except that motor pauses on obstruction detection.

3.4.1.13 dop_EmergencyHandle_Seq Function

This function turns off the motor and waits for the other commands.

3.4.1.14 dop_StartUp Function

Figure 3-22 shows the sequence used to perform the Start-Up of the Door Seek-Home Operation.

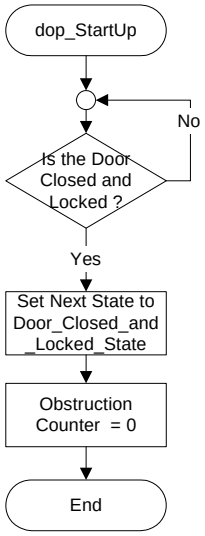

Figure 3-22: dop_StartUp Flowchart

3.4.1.15 dop_SeekHome_Close_Seq Function

Figure 3-23 shows the sequence used to perform a Door Close during a Door Seek-Home Operation.

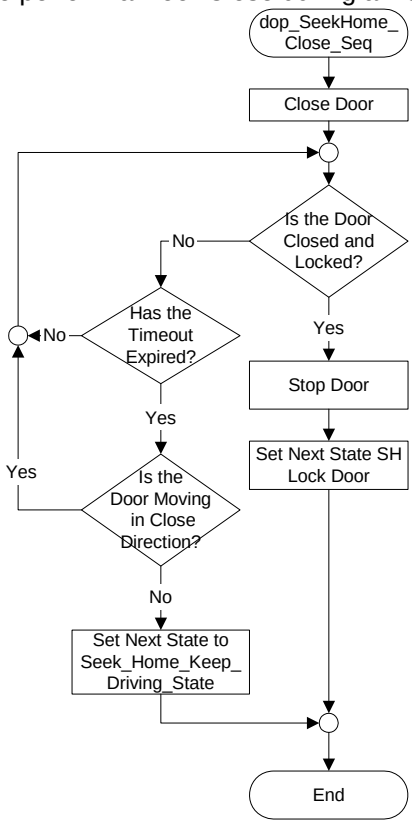

Figure 3-23: dop_SeekHome_Close_Seq Flowchart

3.4.1.16 dop_SeekHome_Door_Stop_Seq Function

Figure 3-24 shows the sequence used to perform a Door Stop during a Door Seek-Home Operation.

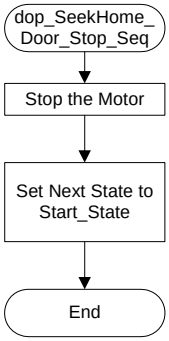

Figure 3-24: dop_SeekHome_Door_Stop_Seq Flowchart

3.4.1.17 dop_SeekHome_Closing_Obstruction_Seq Function

Figure 3-25 shows the sequence used to perform a Door Closing Obstruction during a Door Seek-Home Operation.

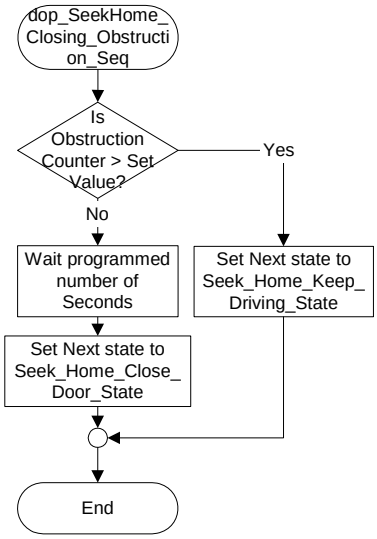

Figure 3-25: dop_SeekHome_Closing_Obstruction_Seq Flowchart

3.4.1.18 dop_MotorDriveTest Function

This function only runs just before the “open door” state, and when both open and close motor power supplies are available. The test is performed by activating one H-Bridge MOSFET at a time and verifying that the normal current limiter is not triggered. If the Normal current limiter is triggered, this function sets event “Motor Drive Circuit (H-Bridge) Error”. The event is reset the next time the test is executed and the normal current limiter is not triggered.

3.4.1.19 DOP_ConfigParameters Function

This function saves the new received configurable parameters or just sends the current parameters. The structured data is stored in a non-volatile memory that is described Table 3-1. The process is described in Figure 3-26.

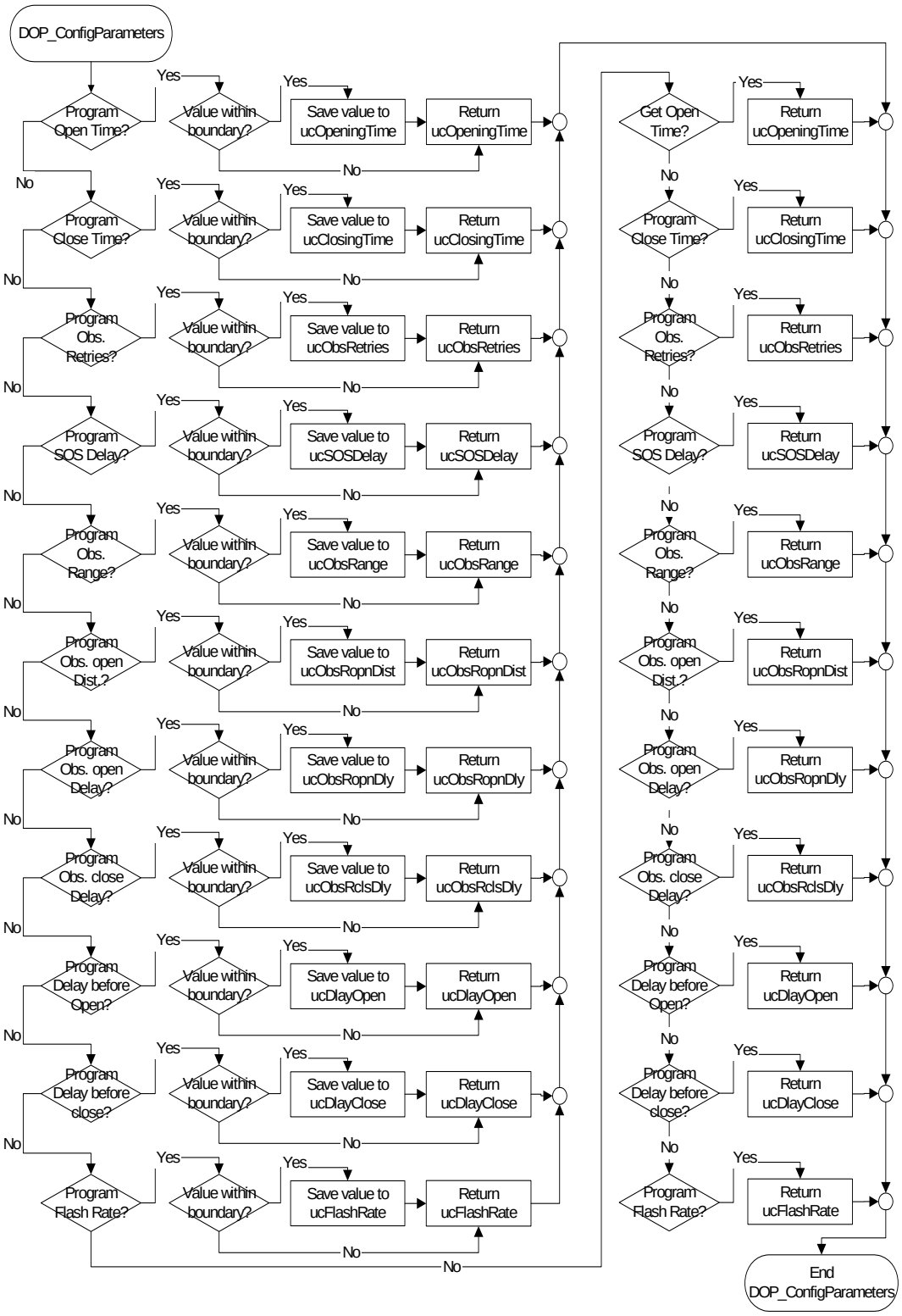

Figure 3-26: DOP_ConfigParameters flowchart

3.4.1.20 DOP_Init Function

This function verifies the integrity of the data ConfigParameters (in Table 3-1) by calling the function CRC_OnRegion function of the CRC Module. It also initializes all variables.

3.4.2 Data Entities Detailed Design

See section 3.4.1 for data description.

APPENDIX A: SPECIFIC PROJECT DESIGN ELEMENT

The following items are defined in the specific project's *Software Applications Configuration Specifications (SACS)*.

A.1: Event Code Number

The event code numbers for the following events are defined in the *SACS* document:

- The code number
- The Event Name

Event Code and Name

Event Code	Event Name
XX	Motor OC Shutdown Circuit Failure
XX	Door Opening Time Longer than Expected
XX	Door Closing Time Longer than Expected
XX	Door Locking Time Longer than Expected
XX	Normal Current Limit Error Detected
XX	High Current Limit Error Detected
XX	PSS Signal Error
XX	Door Closing Obstruction Detected
XX	Door Opening Timeout
XX	Motor Control Module Communication Error
XX	Current Level Selector Error
XX	Motor Drive Circuit (H-Bridge) Error
XX	Motor Rotation with Door Locked